

Phase 5: Project Development Phase

OptiCrop: Smart Agricultural Production Optimization

Role	Name
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Location of Source Code

For compilation, execution safety, and version control best practices, all runnable code is placed directly in the repository's root directory structure rather than being duplicated inside the documentation folder. The main logic resides in two specific sub-folders: `Training/` and `Flask/`.

Source Code Overview

- `Training/model_training.ipynb`: A Jupyter Notebook containing the entire data science pipeline. It includes data loading, K-Means clustering visualization, train-test splitting, scaling, training multiple classifiers, and executing the `pickle.dump()` commands to save the final Logistic Regression model and Standard Scaler.
- `Flask/app.py`: The Python script that initializes the Flask application. It loads the serialized `.pkl` files, defines the routing logic for the 3 web pages, captures user input, applies the scaler, invokes the model, and returns the HTML response.
- `Flask/templates/`: Directory containing `index.html`, `about.html` and `predict.html`.
- `Flask/static/`: Directory containing `style.css` for consistent styling across all web pages.

Dataset Path

The raw dataset utilized for training and analysis is located at:
`Dataset/Crop_recommendation.csv`